CSE 420: Compiler Design

Assignment 2: CLR(1) Parser's Automaton Drawing

Submission Deadline: Saturday, July 26 (In submission box at 4G06)

In this assignment, you will draw the automaton for a Canonical LR(1) , also called CLR(1), parser for a reference grammar. You should define the production rules and alphabet of terminals for the grammar following the process described below. To learn how to construct the automaton of a CLR(1) parser, read Sections 4.7.1 to 4.7.3 of the reference textbook. Draw the automaton exactly in the way shown in the reference textbook. Note that your submission must be a handwritten version, a typed submission will get a 0 out of 10.

Grammar Construction Process (note that ε means empty):

A -> B A C
A -> D E X
B -> the last digit of your student ID
C -> the second last digit of your student ID
D -> the third last digit of your student ID
E -> the fourth last digit of your student ID
Then, if your student ID is odd then add the following rules:
B -> ε
C -> ε
X -> C
Otherwise, add the following rules:
A -> ε
E -> ε
X -> E

An example: Suppose your student ID is 20130165. Then your grammar is as follows

A -> B A C
A -> D E
B -> 5
C -> 6
D -> 1
E -> 0
B -> ε
C -> ε
X -> C

Alphabet of terminals is {5,6,1,0}

Late Submission Policy: Since you get two extra days beyond the regular two weeks' time, no late submission will be accepted for this assignment.